

Bangladeshi License Plate Detection and Recognition with Morphological Operation and Convolution Neural Network

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Abstract—In today's world automatic license plate recognition play an important role in monitoring and organizing vehicles. In this paper, we propose a method of detecting and recognizing the license plates of vehicles in an automatic way in our country. The system can be used to collect toll, in car parking and find stolen vehicles. We have used different image processing techniques like resizing image, image binarization, connected component analysis, image enhancement and noise filtering. Our system is composed of four main modules, such as detection of the license plate area, extraction of license plate. Then, segmentation of characters and words and finally recognition of the characters and words. As Bangladesh Road Transport Authority (BRTA) imposed a common standard for vehicle license plate, the size and aspect ratio of all license plates are same. We have used a threshold value to detect and extract the license plate based on our analysis. Afterward, for character segmentation, we used connected components. Later, we used deep learning tool 'the Convolutional Neural Network' for character recognition. Due to the lack of a standard data set, we have developed a customized dataset of Bangladeshi number plate for the implementation of our method. The accuracy of our proposed system is 95.42%.

Keywords—Vehicle license plates detection, Aspect ratio, Connected Component, Morphological Analysis, and CNN.

I. INTRODUCTION

Bangladesh is a land of 163 million (2016 est.) people. Public transportation is not enough for this large amount of population. The qualities of public transportation also not good enough to satisfy the people. So private transportation is increasing day by day. According to the Bangladesh Road Transport Authority (BRTA), Bangladesh has nearly 3.1 million registered vehicles, and one million of them are in Dhaka. Still, transportation is not adequate; hence there is an overtaking tendency that leads to road accidents and traffic jam. Other important causes are random parking of vehicles, violation of traffic rules and streets occupied by the peddler.

To monitor these vehicles, we need the right amount of manpower that is time-consuming, costly and tedious job. But we can solve this problem by making an automated system that can monitor a vehicle by taking pictures of that vehicle using a high-resolution camera or putting a webcam in the interested area. Automated License Plate Recognition (ALPR) system also can be used to find stolen cars, collecting toll and parking cars in a proper way.

Earlier, there were different types of number plate for different types of vehicles. However, from 2013 BRTA provides the common shape of license plate for every type of vehicles, called Retro Reflective license plate or Digital license plate [1] show in fig 1. The plate has two rows, upper row with alphabetic words and the lower row with numeric digits. In the upper row, the first word is short for a city name, then the second word metro for the metropolitan area. But for the non-metropolitan area, there is just only one word the name of the city and a single Bangla character refers to the category of the vehicle [2]. In the bottom row, there are six digits and these six digits are divided into two groups. First group with two digits and second group contain four digits. The first two digits represent the vehicle class in Bengali numerals and 2nd four digits represent the vehicle number of the vehicle in Bangla numerals. We focused on detection and recognition of Bangladeshi number plate that has been standardized by BRTA.

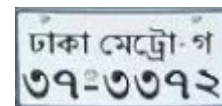


Fig. 1. Retro Reflective License Plate

A lot of researches have been done in the field of vehicle license plate detection and recognition. Similarly, researchers contributed in recognition of vehicle license plate. Ashim Kumar Ghosh et al. [3] proposed a method for

Automatic License Plate Recognition (ALPR) for Bangladeshi Vehicles by using chain code and neural network and the accuracy rate was 84% for license plate extraction and 80% for character recognition.

Kaushik Deb et al. [4] used HSI color model and geometrical properties, and the average accuracy rate is 84.8%.

Our system use different features that were not used combined in the previous work. In this system, we use Morphological Analysis, Aspect ratio of license plate, Boundary detection, and Convolutional neural network for classification. Our System accuracy is 92.78% for license plate area detection, 96.45% for character segmentation and 99.03% for character recognition.

II. LITERATURE REVIEW

Over the last decade, lots of work has been carried out by the researchers for automated license plate detection. They used different approaches mostly color based approaches, region based approaches etc.

The feature of Bangla script called “Matra”, i.e. a line running along the tops of the letters that links them together. where every word is segmented as single connected component that reduces the complexity of segmentation [5], in this paper a technique in digital image processing for finding small parts of an image which match a template image called template matching was used for recognition. [3] In this paper, chain code and the neural network were used. Extraction of the license plate was done using Sobel filter, connected component and morphological operation and for recognition of the characters chain code was used.

Hue-Saturation-Intensity(HSI) color model can be used for finding the candidate region [4] and other approaches like area detection, bounding box, aspect ratio were used to find the region that contain the license plate. In [6] Hough Transform and Voted Block Matching were used for plate extraction.

Edge statistics and morphological operation can be used for the extraction of area [7]. In [8] Dilation morphology and scoring according to the height-width ratio was used for extract of the candidate region and the number of connected components in the region. CNN and SVM were used to classify the region with license plate and without license plate. [9] In this paper, different kind of morphological operation was used to reduce the noise and prepare the image for further process. Canny Edge detection, dilation, erosion and Connected Component Approach are used to extract the license plate and recognition is done using the Template Matching. [10] In this paper Gaussian Filtering and Bayesian Network were used.

Convolutional Neural Network(CNN) is the best tool for recognition. So CNN can be used for character recognition after extracted from the license plate [11]. [12] In this paper, character features was used to extract the license plate. If the position contains some characters, that portion is extracted as license plate.

III. PROPOSED METHOD

The block diagram for the proposed system is shown in the fig 2. It consists of Images Acquisition, Images Preprocessing, License Plate Area Detection, Character Segmentation, and Recognition using CNN.

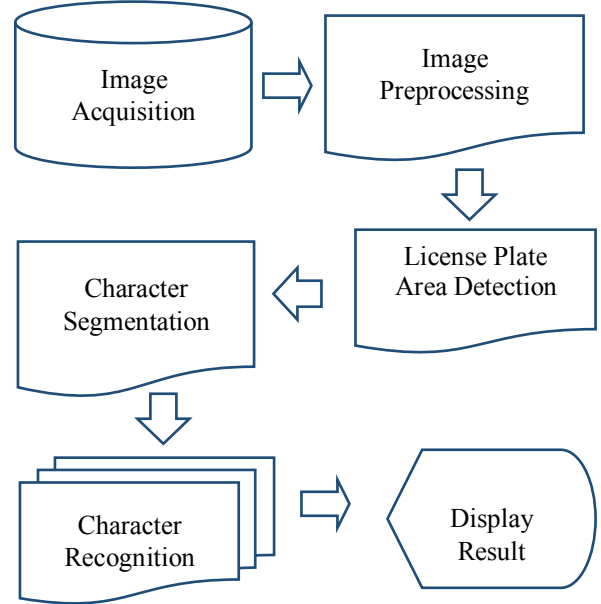


Fig. 2. Proposed System

A. Image Acquisition

In our system, we used a high-resolution camera to capture different vehicle images with the license plate. The images were taken in color mode. The quality of the images was different depending on the surrounding environment. Some of the images were taken where the license plate was displayed in slanted.

B. Image Pre-processing

We used different pre-processing techniques like resizing, noise reduction, and image enhancement etc. At first, the input images were resized at 400x400 pixels. After that we used haze removal techniques to enhance low-light images. It comprises into inverting the low-light images, applying the haze removal algorithm to this images, then again inverting the enhanced image [13] Fig 3(a) and Fig 3(b).



Fig. 3(a). Original image



Fig. 3(b). Enhanced image

We convert the enhanced Images into grayscale fig 4(a) and 4(b) using the equation 1.

$$F(x) = 0.2989 * R + 0.5870 * G + 0.1140 * B \quad (1)$$



Fig. 4(a). RGB image



Fig. 4(b). Grayscale image

After that the noise of grayscale image is removed using the wiener 2-D filter [13] Fig 5(a) and Fig 5(b). Wiener 2 filter approximates the local mean and variance around each pixel using the following equations 2 and 3.

$$\mu = \frac{1}{NM} \sum_{n_1 n_2 \in \eta} a(n_1, n_2) \quad (2)$$

and

$$\sigma^2 = \frac{1}{NM} \sum_{n_1 n_2 \in \eta} a(n_1, n_2) - \mu^2 \quad (3)$$

Where η is the N-by-M local neighborhood of each pixel in the image. Then pixel-wise winner filter is created using equation 4

$$b(n_1, n_2) = \mu + \frac{\sigma^2 - v^2}{\sigma^2} (a(n_1, n_2) - \mu) \quad (4)$$

Where v^2 is the noise variance. If the noise variance is not given, wiener 2 usage the average of all the local estimated variances.



Fig. 5(a). Noise image



Fig. 5(b). Without noise

C. License Plate Area Detection

After preprocessing the images, the next step is detection of the license plate within the whole image. At first, we binarize the grayscale images and converted into label connected components in 2-D binary images. We detect all the connected components, check whether the detected area aspect ratio is between 1.9 and 2.1. If the aspect ratio falls between this ranges, then the area is detected as a license plate, in fact the aspect ratio of license plate is 2. But we chose this ranges because there may be some distortion (Fig 6(a) and Fig 6(b)).

After detecting the license plate we rescale it into 400x400 pixels. The rescaled plate contains some horizontal or vertical lines at the corner of each side. We discard these lines to get only the characters in the license plate by reducing the images pixel size fig 6(b) and 7(a).



Fig. 6(a). Grayscale image

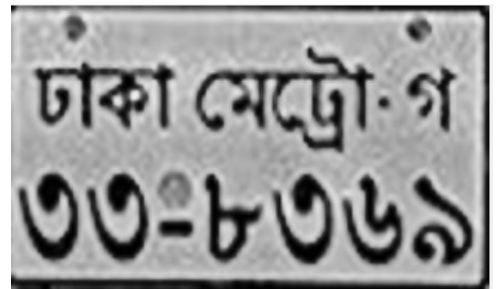


Fig. 6(b). Detected rectangle

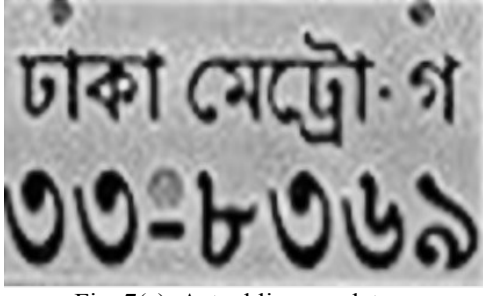


Fig. 7(a). Actual license plate

D. Character Segmentation

Bangladeshi license plate contains characters in two lines. First line contains one or two Bangla words and one Bangla letter. Second line has 6 digits according to Retro Reflective standard of BRTA. All 6 digits are separated from each other. But the alphabets that form a word are connected by using “Matra” (a line connects multiple characters). This whole word connecting with a “Matra” is considered as one connected component.

For character segmentation we used connected component analysis. We first binarized the images, then for each connected component bounding box is formed [15]. We used a minimum threshold value of 20 pixels for heights and widths. Fig 8(a) and Fig 8(b).

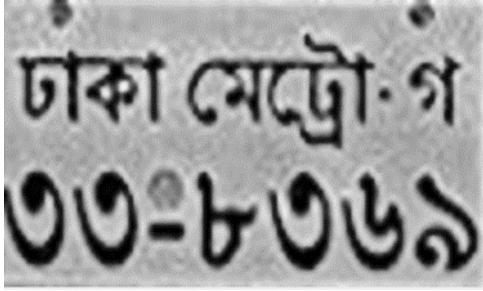


Fig. 8(a). License plate



Fig. 8(b). Segmented characters

E. Recognition

After segmenting characters, the next step is character recognition. For recognition, we use the convolutional neural network. Our network architecture contains three convolution layers; three max pooling layers, three ReLU layers, 16 fully connected layers, one softmax layer and one classification layer.

In our system we use 16 classes. Three for letters “ক”, “খ”, “গ” ten for numerals “০”, “১”, “২”, “৩”, “৪”, “৫”, “৬”, “৭”, “৮”, “৯” and three for “ঢাকা”, “চট্ট”, “মেট্রো”.

Our training phase is shown in Fig 9.

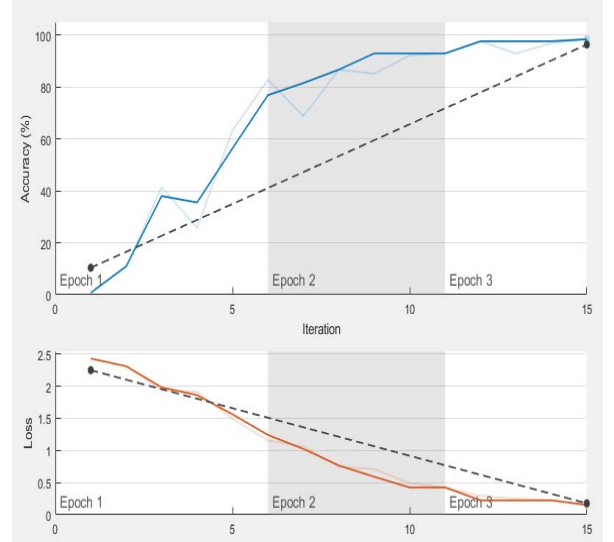


Fig. 9. Training phase

IV. RESULT AND DISCUSSION

For our experiment we use MATLAB17b on Intel core i7 with 8 GB RAM and NVIDIA GeForce 930M (2GB) GPU.

There is no existing dataset for Bangla digits and characters, so we create a customized dataset.

For training and testing the data we have experimented in different ratio. Firstly, 75 for training and 25 for testing. Then 80 for training and 20 for testing but as our data set is small, we get better result for the ratio 85 for training and 15 for testing, as shown in table 1

TABLE I. RATIO OF TRAINING AND TESTING DATA

Ratio of Training and Testing Data	Accuracy
75:25	82.32%
80:20	93.06%
85:15	97.03%

Our proposed system is mainly divided into three parts: License Plate Detection, Character Segmentation, and Character Recognition. The accuracy of each part is shown in table 2 and Fig 9.

TABLE II. MODULE AND ACCURACY

Module	Accuracy
License Plate Detection	93.78%
Character Segmentation	95.45%
Character Recognition	97.03%

Our License Plate Detection accuracy is 93.78%. The accuracy may decrease when the other area also have same aspect ratio as license plate. Our character segmentation accuracy is 95.45% and for the character recognition accuracy is 97.03%. If the characters in the license plate are not connected well, the accuracy may loss.

V. CONCLUSION

In this paper, an attempt has been made to recognize the Bangladeshi license plate using aspect ratio, connected components, and CNN. The result of obtaining from this study is encouraging. However, some issues need to be addressed in this detection. 1) Lack of datasets for Bangla license plate. 2) High-resolution images need to be captured, otherwise characters segmentation detect some broken characters.

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